Week 5 – Payment Transaction Management System

1. Payment Table Design

The Payment table stores payment details for customer orders. Each payment has a unique Payment_ID and is linked to an order using Order_ID.

Column Name	Data Type	Constraint
Payment_ID	INT	Primary Key
Order_ID	INT	Foreign Key, UNIQUE
Payment_Date	DATE	NOT NULL
Payment_Mode	VARCHAR(30)	NOT NULL
Payment_Status	VARCHAR(20)	NOT NULL
Transaction_Amount	DECIMAL(10,2)	NOT NULL, CHECK > 0

The UNIQUE constraint on Order_ID ensures that each order can have at most one payment record. Payment modes include UPI, Credit Card, Debit Card, Net Banking and Cash on Delivery. Payment statuses include Pending, Success and Failed.

2. Updated ER Diagram (Order–Payment Relationship)

ORDERS

Order_ID (Primary Key)

Customer_ID, Order_Date, Total_Amount, Order_Status

One order can have zero or one payment

PAYMENT

Payment_ID (Primary Key)

Order_ID (Unique Foreign Key)

Payment_Date, Payment_Mode, Payment_Status, Transaction_Amount

The relationship between Orders and Payment is one-to-zero-or-one. An order may not have a payment record yet, but each payment must belong to one existing order. The foreign key maintains referential integrity.

3. Sample Customer, Order and Payment Data

Customer table

Customer_ID	Customer_Name	Email
1	Arun	arun@gmail.com
2	Priya	priya@gmail.com
3	Kavin	kavin@gmail.com
4	Divya	divya@gmail.com

Orders table

Order_ID	Customer_ID	Total_Amount	Order_Status
301	1	1550.00	Delivered
302	2	2300.00	Shipped
303	3	2200.00	Pending
304	4	1500.00	Shipped

Payment table

Payment_ID	Order_ID	Payment_Mode	Payment_Status	Amount
501	301	UPI	Success	1550.00
502	302	Credit Card	Success	2300.00
503	303	Debit Card	Failed	2200.00
504	304	Net Banking	Pending	1500.00

Note: These are sample records for documentation. Use your actual database records if they differ.

4. Payment Transaction and Revenue Analysis Reports

A. Payment status report

Payment Status	Number of Payments
Success	2

Failed	1
Pending	1

B. Revenue report

Successful payments are counted as revenue in this example.

Payment Mode	Successful Revenue
UPI	₹1,550.00
Credit Card	₹2,300.00
Total Revenue	₹3,850.00

C. Transaction analysis

- Total payment records: 4
- Successful payments: 2
- Failed payments: 1
- Pending payments: 1
- Total successful revenue: ₹3,850.00
- Average transaction amount: ₹1,887.50

These reports help identify successful transactions, pending payments, failed payments and revenue collected through different payment methods.